25900550 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPES201 - UNDERSTANDING HUMAN BODY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Mitochondria are abundant in cells that perform _____. [U] / [CO1]

   a). Secretion b). Excretion

   c). Active transport d). Energy production

2. Which bone articulates with the glenoid cavity to form the shoulder joint? [U] / [CO2]

   a). Radius b). Ulna

   c). Clavical d). Humerus

3. Which molecule provides the energy for muscle contraction? [A] / [CO1]

   a). Oxygen b). ATP

   c). Glucose d). Lactic acid

4. Which type of neuron is found in the spinal cord and connects sensory and motor neurons? [U] / [CO1]

a). Relay neuron b). Sensory neuron

c). Motor neuron d). Interneuron

5. What is the basic structural and functional unit of the nervous system? [U] / [CO1]

a). Glial cell b). Neuron

c). Nerve fiber d). Synapse

6. Which part of the brain controls the rate of breathing? [U] / [CO2]

a). Cerebellum b). Medulla oblongata

c). Cerebrum d). Thalamus

7. Which term refers to moving a limb away from the midline of the body? [An] / [CO2]

a). Adduction b). Flexion

c). Abduction d). Circumdation

8. Which muscle fiber has the highest mitochondrial content? [K] / [CO1]

a). Type I b). Type IIa

c). Type IIb d). None

9. In double circulation, blood passes through the heart [U] / [CO2]

a). Once b). Twice

c). Three times d). Four times

10. The nervous system is broadly divided into how many main parts? [U] / [CO1]

a). One b). Two

c). Three d). Four

[1x10 = 10]

Part B

SHORT ANSWER QUESTIONS

Answer any **10** questions

Each question carries **2** marks

11. What is blood classified as? [U] / [CO1]

12. What is the function of the spinous process? [An] / [CO6]

13. Define muscle fiber fatigue. [U] / [CO1]

14. What is the role of the pituitary gland? [U] / [CO2]

15. How does the urinary system remove nitrogenous wastes? [A] / [CO2]

16. Give two examples of anatomical directional terms. [U] / [CO5]

17. Give two general features of the scapula. [U] / [CO2]

18. State the function of actin and myosin in muscle contraction. [A] / [CO2]

19. Give two examples of mixed nerves. [U] / [CO1]

20. What is peristalsis? [U] / [CO2]

21. Define the term 'caudal'. [U] / [CO5]

22. What is a ball and socket joint? Give an example. [U] / [CO2]

[2x10 = 20]

Part C

SHORT ESSAY QUESTION

Answer any **4** questions

Each question carries **5** marks

23. Discuss the functional classification of muscles with examples. [U] / [CO1]

24. Explain the structure and functions of the heart with a neat diagram. [U] / [CO2]

25. Explain the role of the spinal cord in reflex action and communication. [U] / [CO1]

26. Discuss the structural features and functions of neutrophils. [S] / [CO3]

27. Explain bone remodeling and factors that influence it [U] / [CO2]

28. Explain irritability in muscles and its importance for survival. [U] / [CO1]

[5x4 = 20]